CAMBRIDGE IGCSE CHEMISTRY 0620

Topic 8: The Periodic Table

Premium Answer Booklet

10 questions	2020-2025 Paper 4	Premium readable layout

Rebuilt for easier reading with structured tables, clear answer sections and highlighted workings. Use beside the matching topical question bank.

Contents

Question 01

12 marks

2020 Feb/March Paper 42 Question 4

Main topic: Topic 8: The Periodic Table
Cross-topic links: Topic 7: Acids, bases and salts, Topic 9: Metals, Topic 6: Chemical reactions
Answer source label: 0620_m20_ms_42.pdf

4(a)	• Haber (process) (1) • ammonia (1)
4(b)(i)	• green
4(b)(ii)	• $Fe^{2+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_2(s)$ • $Fe(OH)_2$ (as only product) (1) • Fe^{2+} and $2OH^-$ (as reactants) (1) • state symbols (1)
4(c)(i)	• oxidising agent
4(c)(ii)	• presence of an acid
4(c)(iii)	• lose an electron
4(c)(iv)	• colourless
4(d)	• 3+ • 3+

Easy explanation / reminder
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 02 — 16 marks

2020 Oct/Nov Paper 43 Question 5

Main topic: Topic 8: The Periodic Table
Cross-topic links: Topic 4: Electrochemistry
Answer source label: 0620_w20_ms_43.pdf

5(a)	• become more reactive down the group opposite reasoning also allowed (1)
5(b)(i)	• one mark each for any two of: • floats • dissolves / disappears / melts • moves • bubbles / fizzes / effervesces • lilac flame
5(b)(ii)	• 2K + 2H2O -> 2KOH + H2 • all formulae (1) • equation fully correct (1)
5(c)(i)	• Cl2 + 2KI -> 2KCl + I2 • OR Cl2 + 2I- -> 2Cl - + I2 • all formulae (1) • equation fully correct (1)
5(c)(ii)	• brown / black
5(d)(i)	• breakdown by (the passage of) electricity (1) • of an ionic compound in molten or aqueous (state) (1)
5(d)(ii)	• heat until it melts / heat to or above melting point
5(d)(iii)	• Na+ + e -> Na
5(e)(i)	• one mark for each of any two from: • (chromium has) high melting point opposite reasoning also allowed • (chromium forms) coloured ions / coloured compounds opposite reasoning also allowed • (chromium has) variable valency / variable oxidation state / variable oxidation number opposite reasoning also allowed • catalytic behaviour opposite reasoning also allowed • opposite reasoning also allowed ALLOW group 1 or sodium if stated • no colour or white or colourless ions or compounds • fixed valency / +1 charge only or one oxidation state / forms one chloride • low melting point • doesn't behave as a catalyst
5(e)(ii)	• one mark for each of any two from: • (chromium / sodium) conducts electricity • (chromium / sodium) compounds are soluble (in water) • (chromium / sodium) form hydrated salts / form hydrated compounds

Easy explanation / reminder
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 03	22 marks

2021 Oct/Nov Paper 42 Question 5

Main topic: Topic 8: The Periodic Table
Answer source label: 0620_w21_ms_42.pdf

5(a)	• metallic (1) • (lattice of) positive ions (1) • sea of / delocalised / mobile electrons (1) • attraction between positive ions and electrons (1)
5(b)(i)	• malleability / malleable
5(b)(ii)	• (particles) slide (over each other)
5(b)(iii)	• unreactive
5(c)	• high(er) density • high(er) melting points
5(d)(i)	• Na - yellow (1) • K - lilac (1) • colourless solution (1) • clear idea of increased reactivity in both reactions (1)
5(d)(ii)	• hydrogen
5(d)(iii)	• potassium hydroxide
5(d)(iv)	• any number in the range 7< pH ■ 14
5(d)(v)	• $4Na + O_2 \rightarrow 2Na_2O$ • Na_2O (1) correct equation (1)
5(e)(i)	• $N_2 + 3H_2 \rightarrow 2NH_3$ • species (1) correct equation (1)
5(e)(ii)	• 450 °C (1) • 200 atm (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.

END OF ANSWER

Question 04	11 marks

2022 May/June Paper 41 Question 7

Main topic: Topic 8: The Periodic Table
Answer source label: 0620_s22_ms_41.pdf

7(a)(i)	• from left to right • caesium -> rubidium-> potassium ->sodium -> lithium
7(a)(ii)	• caesium hydroxide
7(b)	• Group I element is less strong / not strong • opposite reasoning also allowed • OR • Group I element has low(er) density opposite reasoning also allowed • OR • Group I element is soft(er) opposite reasoning also allowed
7(c)	• solid
7(d)(i)	• colourless (1) • orange / brown / yellow (1)
7(d)(ii)	• Br - (1) • loses electron(s) (1)
7(e)	• 432(1) • 436(1) • - 4(1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.

END OF ANSWER

Question 05

16 marks

2022 May/June Paper 43 Question 6

Main topic: Topic 8: The Periodic Table
Answer source label: 0620_s22_ms_43.pdf

6(a)(i)	• Group 1 metals do not show catalytic behaviour • Group 1 have fixed oxidation states
6(a)(ii)	• any 2 observations from: • ■ • moves / floats • ■ • dissolves / disappears • ■ • bubbles / effervescence / fizzes • ■ • lilac flame • ■ • explodes • ■ • melts / forms a spherical shape • $2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$ • KOH or H_2 product (1) in equation • fully correct equation (1) • state symbols (1)
6(b)	• transition elements have high(er) melting point (1) • transition elements have high(er) density (1)
6(c)(i)	• colourless (1) • brown (1)
6(c)(ii)	• redox
6(c)(iii)	• Br_2 (1) • (bromine) is reduced (1)
6(d)	• two ticks in first row (1) • three crosses in the other three boxes (1)

Easy explanation / reminder
• Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.

END OF ANSWER

Question 06

18 marks

2023 Feb/March Paper 42 Question 4

Main topic: Topic 8: The Periodic Table
Cross-topic links: Topic 3: Stoichiometry, Topic 9: Metals
Answer source label: 0620_m23_ms_42.pdf

4(a)(i)	• zinc
4(a)(ii)	• alloy
4(b)(i)	• ductility
4(b)(ii)	• electrons
4(c)(i)	• high melting point
4(c)(ii)	• (act as) catalysts
4(d)(i)	• water(s) of crystallisation
4(d)(ii)	• blue
4(d)(iii)	• CuSO4■5H2O • CuSO4 (1) • ■5H2O (1)
4(e)(i)	• basic
4(e)(ii)	• the oxidation number of copper is +2
4(e)(iii)	• Mr of Cu(NO3)2 = 188 (1) • = 0.0200 -> = 0.0200 -> 188 = 3.76 g (1) • February/March 2023
4(e)(iv)	• moles of gas formed = 0.0200 -> 5 / 2 = 0.05(00) (1) • volume = -> 24.0 = 0.05(00) -> 24.0 = 1.2(0) (1)
4(e)(v)	• 2Al + 3CuO -> Al2O3 + 3Cu • correct products (1) • rest of equation correct (1)

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than
• activation energy.

END OF ANSWER

Question 07

18 marks

2024 Feb/March Paper 42 Question 2

Main topic: Topic 8: The Periodic Table
Cross-topic links: Topic 2: Atoms, elements and compounds
Answer source label: 0620_m24_ms_42.pdf

2(a)	• fluorine
2(b)	• red-brown AND liquid
2(c)	• Ts • 7
2(d)(i)	• pair of electrons • electron(s) shared between two atoms
2(d)(ii)	• iodide / astatide / tenesside
2(d)(iii)	• bromine is more reactive than iodine / astatine / tenessine
2(e)	• cobalt(II) chloride • anhydrous
2(f)	• 8 crosses in third shell of Ca • 7 dots and 1 cross in third shell of both Cl • ‘2+' charge on Ca on correct answer line AND ‘-‘ charge on both Cl ions on correct answer line • February/March 2024
2(g)(i)	• lead(II) nitrate
2(g)(ii)	• Pb2+(aq) + 2Cl -(aq) -> PbCl2(s) • PbCl2 as only product • Pb2+ + 2Cl - as only reactants • correct state symbols
2(g)(iii)	• silver chloride

Easy explanation / reminder
• Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation
• before converting to the final answer.
• Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be
• reduced by carbon/CO.

END OF ANSWER

Question 08

13 marks

2025 Feb/March Paper 42 Question 3

Main topic: Topic 8: The Periodic Table
Answer source label: 0620_m25_ms_42.pdf

3(a)	• VII
3(b)	• No separate point listed.
3(c)(i)	• tennessine / Ts / Ts2
3(c)(ii)	• fluorine / F / F2
3(d)	• halide(s)
3(e)	• No separate point listed.
3(f)	• krypton
3(g)(i)	• Cl 2 + 2I- -> 2Cl - + I2 • Cl- • correct ionic equation
3(g)(ii)	• I- loses e - • Cl2 gains e -
3(h)	• grey-black • solid • February/March 2025

Easy explanation / reminder
• Structure/trend tip: link properties to particles, bonding and outer-shell electrons. Do not just state the trend; explain it.

END OF ANSWER

Question 09

12 marks

2025 May/June Paper 41 Question 6

Main topic: Topic 8: The Periodic Table
Cross-topic links: Topic 5: Chemical energetics
Answer source label: 0620_s25_ms_41.pdf

6(a)	• halogen(s)
6(b)	• fluorine / F_2
6(c)	• solid • grey-black
6(d)(i)	• $Br_2 + 2I^- \rightarrow 2Br^- + I_2$ • I^- as a reactant and Br^- as a product • equation fully correct
6(d)(ii)	• 343 • 350 • -7
6(e)(i)	• any two from:- • solid dissolves / solid disappears • bubbling / effervescence / fizzing • melts / forms a ball • floats • moves
6(e)(ii)	• blue

Easy explanation / reminder
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
• Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.

END OF ANSWER

Question 10

15 marks

2025 May/June Paper 43 Question 6

Main topic: Topic 8: The Periodic Table
Answer source label: 0620_s25_ms_43.pdf

6(a)	• alkali metals
6(b)	• lithium
6(c)(i)	• Any two: • solid dissolves / disappears • fizzing/ bubbles / effervescence • solid floats • solid moves (on surface)
6(c)(ii)	• 2Li + 2H2O -> 2LiOH + H2 • LiOH as product • correct equation
6(d)	• softer • lower density
6(e)(i)	• gas • pale yellow-green
6(e)(ii)	• Cl2 + 2Br- -> 2Cl - + Br2 • Br- as a reactant and Cl - as a product • correct equation
6(e)(iii)	• energy required in breaking bonds • = 150 + 242 = (+) 392 • energy released when bonds are formed • = 2 -> 218 = (+) 436 • enthalpy change, ΔH • = 392 - 436 = - 44

Easy explanation / reminder
• Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.
• Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.

END OF ANSWER